

Highlights

Execution-Aware Multi-Market Value Stacking for Behind-the-Meter Campus EV Charging and Battery Storage

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- Executed first-step accounting couples campus EV service, retail charges, and wholesale value.
- Fixed DA commitments and signed RT adjustments form a consistent two-settlement model.
- Rolling 24-h policies are validated against a full billing-period Perfect benchmark.
- All 5,820 Rolling solves and all 19 physical, financial, and solver checks pass.
- The oracle benchmark exceeds the best 24-h policy by 14.38% in June 2025.

Execution-Aware Multi-Market Value Stacking for Behind-the-Meter Campus EV Charging and Battery Storage

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Abstract

Campus electric-vehicle (EV) charging and battery energy storage systems (BESS) can jointly supply behind-the-meter (BTM) flexibility, but gross wholesale revenue is not a sufficient performance metric because the same dispatch changes retail energy and demand charges. This paper develops an execution-aware mixed-integer model predictive control framework that co-optimizes unidirectional workplace EV charging and a bidirectional BESS across retail time-of-use energy, peak-demand, and non-coincident demand charges; EV user payments; day-ahead (DA) and real-time (RT) energy; and four ancillary-service products. The framework closes three gaps in existing value-stacking studies: it prevents double counting of EV and BESS flexibility through a shared physical deliverability envelope, preserves irreversible DA commitments through signed RT adjustments, and evaluates forecast and horizon value from executed first-step dispatch against a full billing-period oracle. A June 2025 UC San Diego case compares shrinking- and rolling-horizon 24-h policies under Perfect and Persistence EV information. The full-period oracle yields \$12,565.15 and exceeds the best 24-h policy by \$1,579.73 (14.38%). Perfect information over a truncated horizon is not a full-period upper bound because BESS state, demand thresholds, and DA commitments are path dependent. All 5,820 Rolling DA/RT solves are accepted at a 1% relative MIP gap, with no time-limit event and physical and settlement residuals below prescribed tolerances. The resulting contribution is a market-timeline-consistent and independently auditable methodology for quantifying the net, rather than gross, value of BTM EV-BESS participation.

Keywords: Electric vehicle charging, Battery energy storage, Model predictive control, Value stacking, Ancillary services, Demand charges, Behind-the-meter resources

1. Introduction

FERC Order No. 2222 seeks to reduce barriers to distributed-energy-resource (DER) participation in organized wholesale markets and explicitly includes storage, solar, controllable demand, EVs, and charging equipment among candidate resources [1]. For a campus or commercial site, however, market access does not by itself establish profitability. A resource can earn wholesale energy or ancillary-service (AS) revenue while simultaneously increasing retail time-of-use (TOU) cost or establishing a higher monthly demand peak.

Workplace EV charging adds a second complication. Without vehicle-to-grid operation, EVs are not physical storage. Their flexibility is the ability to shift nonnegative charging power while each vehicle is connected and still satisfy its departure-energy requirement. A BESS is bidirectional but limited by power, state of charge (SOC), efficiency, and throughput. When both assets share a meter, their local and market-facing actions must be co-optimized rather than valued independently.

This paper addresses that coordination problem for a measured UC San Diego workplace-charging portfolio. The same optimization contains EV service, BESS physics, a signed BTM meter, retail energy and demand charges, DA/RT energy settlement, and regulation-up, regulation-down, spinning-reserve, and non-spinning-reserve capacity. CAISO identifies these four AS products and requires certification and telemetry for participating generation and load [2]. The model is an operational and accounting abstraction; it is not a substitute for a complete CAISO registration or performance-settlement study.

The central methodological choice is *execution-aware accounting*. Each MPC call produces a future plan, but only the first 15-min decision is implemented. SOC, EV energy, demand thresholds, active DA commitments, and revenue accounts are then updated from that executed quantity. This avoids reporting profits from future plan entries that are overwritten by later MPC calls.

The scientific contributions are:

1. **A non-double-counting multi-resource deliverability model.** Unidirectional EV load modulation and bidirectional BESS power share one signed retail meter, one energy position, and one AS capability envelope. Market capacity cannot be sold independently by each account-

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ing channel.

2. **A market-timeline-consistent execution layer.** Submitted DA energy and AS quantities become immutable RT parameters; signed RT variables represent increments or decrements, and the actual physical obligation is their sum. This replaces an artificial DA-tracking penalty with the economic logic of two settlement.
3. **A billing-period validation benchmark for economic MPC.** Executed 15-min dispatch is compared with a full-period Perfect oracle under common physical and boundary conditions. This exposes horizon value separately from forecast value and provides a formal explanation for non-monotone Perfect/Persistence rankings in truncated MPC.
4. **A complete net-value audit.** Wholesale energy and capacity revenue is reconciled with signed TOU settlement, incremental monthly demand charges, EV user payments, EV service, device states, and interval-level feasibility.

The current paper reports the formally validated June 2025 EV-BESS base case. Measured building-load and PV series are retained for a parallel passive-DER sensitivity case, described mathematically in Section 4.4, but their numerical results are not mixed with the validated base-case claims.

2. Related work and research gap

2.1. EV charging flexibility and aggregators

Classical smart-charging studies establish that EV load can be shifted to reduce system cost or grid impacts while respecting availability and energy requirements [3–6]. Aggregation models translate heterogeneous session constraints into a tractable fleet capability [7, 8]. AS studies often assume V2G or bidirectional vehicle power [9, 10]; the present work instead treats EVs as strictly unidirectional loads.

Two recent UCSD studies are the closest EV-method predecessors. Chen et al. [11] optimized workplace EV flexibility for demand-response participation with TOU and demand charges. Chen et al. [12] added real-time feedback under EV forecast uncertainty. The current paper extends that line to co-located BESS operation, wholesale DA/RT energy, four AS products, and executed two-settlement accounting.

2.2. MPC for charging stations and multi-service storage

MPC has been demonstrated for charging stations and grid-connected microgrids [13, 14]. PV-BESS charging-station studies have used explicit or robust-explicit MPC for uncertainty and carbon objectives [15, 16]. These studies show that EV-PV-BESS coordination and MPC are established concepts; neither is claimed as the novelty here.

BTM storage value stacking combines local and grid services but creates competing uses of limited power and energy [17–19]. Santosuoso et al. [20] jointly considered renewable energy, storage, energy trading, AS, and stochastic economic MPC. Warmerdam et al. [21] coordinated PV, EVs, heat pumps, and BESS for reserve provision. These works motivate a precise novelty boundary.

2.3. Research gap and novelty

Table 1 positions the contribution against the closest application classes. Prior work provides strong treatments of EV smart charging, BTM storage, PV-EV-BESS coordination, or energy/AS trading, but the intersection of retail demand-charge state, US-style DA/RT settlement, non-double-counted EV-BESS deliverability, executed first-step accounting, and a billing-period oracle remains unresolved. The paper addresses that intersection. It does not claim a new generic MPC solver or a new forecast; the methodological novelty is the coupling and validation structure needed to make multi-market BTM value claims physically and financially defensible.

3. System, data, and operating sequence

3.1. BTM portfolio

Figure 1 shows the system boundary. EVs and the BESS are active decision resources in the validated base case. The site meter connects the physical dispatch to retail charges and wholesale deliverability. Building load and PV are passive exogenous additions in the planned sensitivity case.

The EV input is an aggregate representation of all modeled UCSD charging sessions rather than a station-by-station electrical topology. Each session retains arrival, departure, energy, and charger-limit constraints. Market prices, tariff schedules, and EV information are aligned to a 15-min time index.

3.2. DA/RT and MPC sequence

Figure 2 separates financial commitment from physical execution. DA quantities are solved before operation and become fixed parameters in RT. Every 15 min, RT optimization updates the EV forecast and physical states, solves the active horizon, and implements one step.

Two 24-h implementations are retained. Shrinking MPC optimizes the remaining service window, so its active horizon shortens as the day progresses. Rolling MPC advances a nominal 24-h window after each execution step. Both use the same device, meter, wholesale, and financial equations.

4. Optimization model

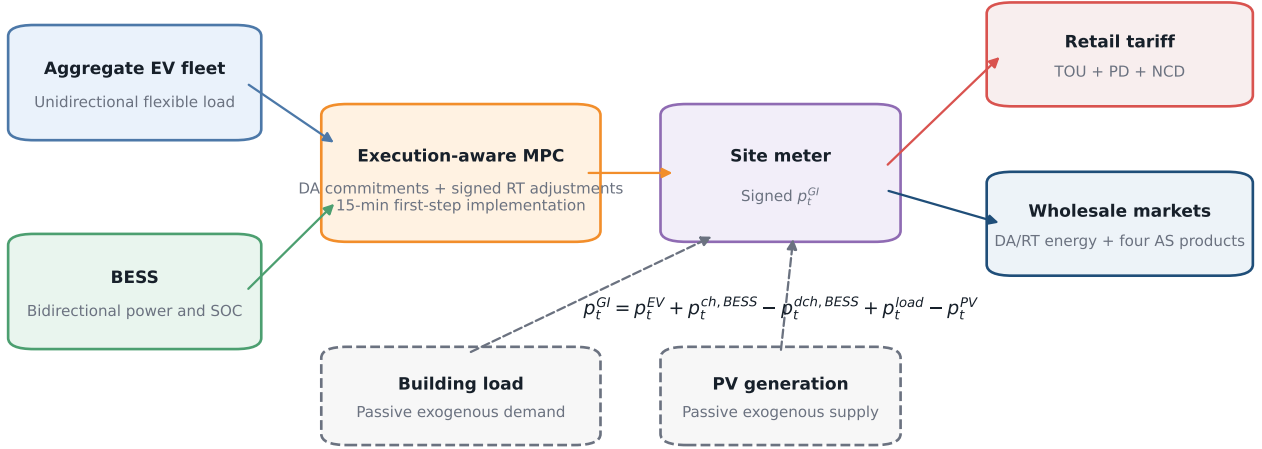
4.1. Sets, variables, and sign conventions

Let $t \in \mathcal{T}$ index 15-min intervals, $i \in \mathcal{I}$ index EV sessions, and $x \in \mathcal{K} = \{\text{RU}, \text{RD}, \text{SP}, \text{NSP}\}$ index AS products. Positive p_t^{GI} denotes site import and negative p_t^{GI} denotes export. Positive market energy p_t^{actual} denotes upward delivery, while negative values denote absorption. The positive-part operator is $[z]_+ = \max(z, 0)$.

Table 1: Positioning relative to closely related application studies. A check mark indicates that the element is central to the cited formulation or validation.

Study	Uni-dir. EV service	BTM demand charges	Energy + AS	DA/RT commitment logic	Full-period oracle	Primary distinction
Chen et al. [11, 12]	✓	✓	–	–	–	Workplace EV demand response under forecast uncertainty.
Cabrera-Tobar et al. [15, 16]	✓	–	–	–	–	Experimental PV-EV-BESS robust/explicit MPC.
Santoso et al. [20]	–	–	✓	✓	–	Stochastic renewable-BESS energy and AS trading.
Id omar et al. [19]	–	✓	✓	–	–	Experimental BTM BESS local/grid service stacking.
Warmerdam et al. [21]	✓	–	✓	–	–	PV, flexible loads, and BESS for aFRR.
This work	✓	✓	✓	✓	✓	Executed US retail-DA/RT-AS value with a shared EV-BESS envelope.

COORDINATED BTM PORTFOLIO AND MARKET INTERFACES



Dashed resources are the parallel passive-DER sensitivity case; they are not decision variables.

Figure 1: Campus BTM architecture. Solid boxes are active in the validated June EV-BESS base case. Dashed building-load and PV boxes define the parallel passive-DER sensitivity case and do not introduce new control variables.

Table 2: Core variables and parameters.

Symbol	Definition
p_t^{EV}, p_t^{EV}	EV-session charging power and fleet total; both are nonnegative.
$B_t^{EV}, p_{t,max}^{EV}$	EV baseline power and connected-charging capability.
$p_t^{ch,BESS}, p_t^{dch,BESS}, s_t$	BESS charge, discharge, and SOC.
$p_t^{DA}, p_t^{RT}, p_t^{actual}$	fixed DA energy, signed RT adjustment, and actual energy position.
$c_t^{DA}, c_t^{RT}, c_t^{actual}$	fixed DA AS, signed RT adjustment, and actual nonnegative AS capacity.
$M_{q,hist}^k$	executed billing-period threshold entering MPC call $k, q \in \{PD, NCD\}$.
C_t^{TOU}, C_t^q	retail energy and demand-charge rates.
$\lambda_t^{DA}, \lambda_t^{RT}, \rho_t^{x,s}$	DA/RT energy and AS capacity prices.

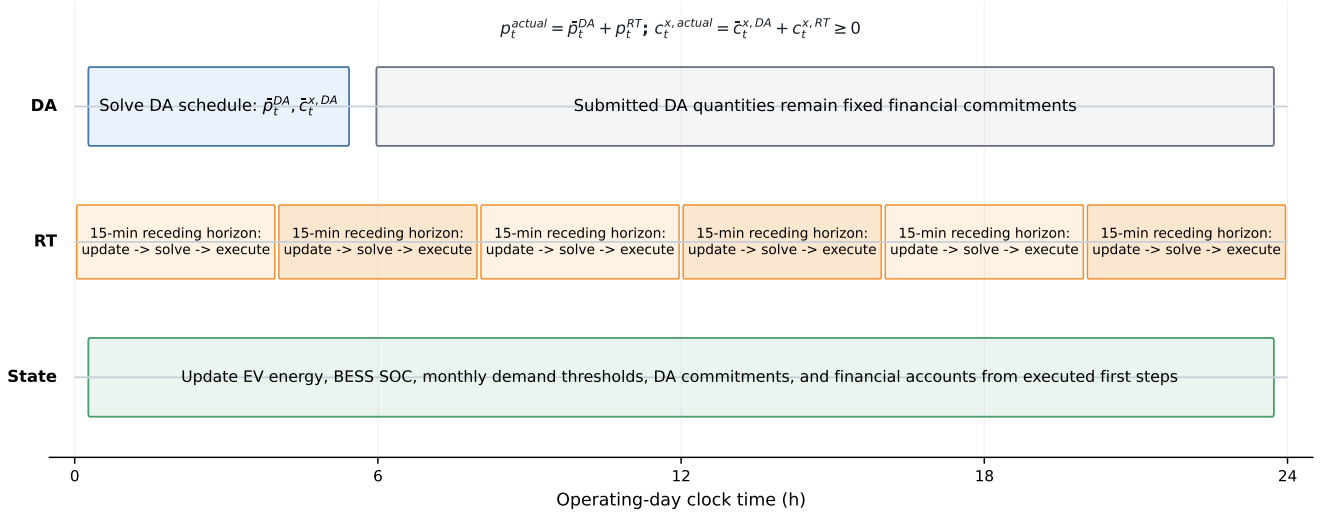


Figure 2: DA commitment, RT adjustment, and first-step MPC execution. Submitted DA quantities cannot be undone; RT variables change the actual position and are settled at RT prices.

4.2. EV service

For availability $b_{t,i} \in \{0, 1\}$, charger rating $\bar{P}_i$, charging efficiency η_i , and required energy E_i^{req} ,

$$0 \leq p_{t,i}^{\text{EV}} \leq \bar{P}_i b_{t,i}, \quad (1)$$

$$e_{t,i} = e_{t-1,i} + \eta_i p_{t,i}^{\text{EV}} \Delta t, \quad (2)$$

$$e_{d,i} = E_i^{\text{req}}, \quad p_t^{\text{EV}} = \sum_i p_{t,i}^{\text{EV}}. \quad (3)$$

The equality in (3) prevents over- or under-service. EV market flexibility is a deviation from B_t^{EV} , not physical discharge:

$$p_t^{\text{EV}} - B_t^{\text{EV}} = (p_t^{\text{ch,EV,WM}} - p_t^{\text{dch,EV,WM}}) + (p_t^{\text{ch,EV,NWM}} - p_t^{\text{dch,EV,NWM}}). \quad (4)$$

Here $p_t^{\text{dch,EV}}$ means a reduction in EV load relative to baseline. It never implies power export from a vehicle.

4.3. BESS physics

Let C^{BESS} be energy capacity, P^{BESS} power rating, and γ the round-trip efficiency parameter implemented through a square-root split:

$$s_{t+1} = s_t + \frac{\Delta t}{C^{\text{BESS}}} \left(\sqrt{\gamma} p_t^{\text{ch,BESS}} - \frac{p_t^{\text{dch,BESS}}}{\sqrt{\gamma}} \right), \quad (5)$$

$$s^{\min} \leq s_t \leq s^{\max}, \quad (6)$$

$$0 \leq p_t^{\text{ch,BESS}} \leq P^{\text{BESS}} u_t^{\text{ch}}, \quad (7)$$

$$0 \leq p_t^{\text{dch,BESS}} \leq P^{\text{BESS}} u_t^{\text{dch}}, \quad (8)$$

$$u_t^{\text{ch}} + u_t^{\text{dch}} \leq 1. \quad (9)$$

A daily throughput guard is imposed separately for every calendar day intersecting the active horizon. SOC carries between

days; only the final evaluation-period SOC is fixed:

$$\Delta t \sum_{t \in T_d} (p_t^{\text{ch,BESS}} + p_t^{\text{dch,BESS}}) \leq 2C^{\text{BESS}}(s^{\max} - s^{\min}), \quad (10)$$

$$s_{T+1} = s^{\text{terminal}}. \quad (11)$$

4.4. Meter balance and passive DER extension

The general signed meter balance is

$$p_t^{\text{GI}} = p_t^{\text{EV}} + p_t^{\text{ch,BESS}} - p_t^{\text{dch,BESS}} + p_t^{\text{load}} - p_t^{\text{PV}}. \quad (12)$$

The validated base case sets $p_t^{\text{load}} = p_t^{\text{PV}} = 0$. In the parallel PV/building sensitivity, the two terms are measured or forecast exogenous parameters. They are passive supply and demand, not new control variables. No import/export split is added; the signed meter topology and all other constraints remain unchanged.

4.5. DA/RT energy and AS settlement

Let $\bar{p}_t^{\text{DA}}$ and $\bar{c}_t^{x,\text{DA}}$ be submitted quantities and m_t an activation mask. In RT,

$$p_t^{\text{DA}} = m_t \bar{p}_t^{\text{DA}}, \quad p_t^{\text{actual}} = p_t^{\text{DA}} + p_t^{\text{RT}}, \quad (13)$$

$$c_t^{x,\text{DA}} = m_t \bar{c}_t^{x,\text{DA}}, \quad c_t^{x,\text{actual}} = c_t^{x,\text{DA}} + c_t^{x,\text{RT}} \geq 0. \quad (14)$$

The RT variables are signed. Hence a negative $c_t^{x,\text{RT}}$ can reduce an AS award that cannot be fully delivered, but the actual provided quantity cannot be negative.

With $P_{t,\max}^{\text{EV}} = \sum_i \bar{P}_i b_{t,i}$, the common capability bounds are

$$\bar{P}_t^{\text{up}} = P^{\text{BESS}} + B_t^{\text{EV}}, \quad (15)$$

$$\bar{P}_t^{\text{down}} = \max\{P^{\text{BESS}} - B_t^{\text{EV}} + P_{t,\max}^{\text{EV}}, 0\}. \quad (16)$$

The nonnegative clipping in (16) is necessary when baseline EV power exceeds the instantaneous BESS-plus-connected-EV absorption capability. Actual energy and actual AS share these bounds:

$$[p_t^{\text{actual}}]_+ + c_t^{\text{RU,actual}} + c_t^{\text{SP,actual}} + c_t^{\text{NSP,actual}} \leq \bar{P}_t^{\text{up}}, \quad (17)$$

$$[-p_t^{\text{actual}}]_+ + c_t^{\text{RD,actual}} \leq \bar{P}_t^{\text{down}}. \quad (18)$$

The implementation also bounds p_t^{DA} and p_t^{RT} individually by the same interval capability limits.

Expected AS deployment gives the market-facing power

$$q_t^{\text{WM}} = p_t^{\text{actual}} + \alpha_t^{\text{RU}} c_t^{\text{RU,actual}} - \alpha_t^{\text{RD}} c_t^{\text{RD,actual}} + \alpha_t^{\text{SP}} c_t^{\text{SP,actual}} + \alpha_t^{\text{NSP}} c_t^{\text{NSP,actual}}. \quad (19)$$

Auxiliary net-flow variables satisfy

$$p_t^{\text{dch,net}} = [q_t^{\text{WM}}]_+, \quad (20)$$

$$p_t^{\text{ch,net}} = [-q_t^{\text{WM}}]_+, \quad (21)$$

$$p_t^{\text{ch,net}} - p_t^{\text{dch,net}} = p_t^{\text{ch,BESS,WM}} - p_t^{\text{dch,BESS,WM}} + p_t^{\text{ch,EV,WM}} - p_t^{\text{dch,EV,WM}}. \quad (22)$$

Direction binaries linearize (20)–(21). This is the role of big- M in the current model. The former big- M penalty for following the DA schedule is removed.

The conservative BESS duration guard for assumed duration T^{DU} is

$$s_{t+1} \geq s^{\min} + \frac{T^{\text{DU}}}{C^{\text{BESS}} \sqrt{\gamma}} \left(c_t^{\text{RU,actual}} + c_t^{\text{SP,actual}} + c_t^{\text{NSP,actual}} \right), \quad (23)$$

$$s_{t+1} \leq s^{\max} - \frac{T^{\text{DU}} \sqrt{\gamma}}{C^{\text{BESS}}} c_t^{\text{RD,actual}}. \quad (24)$$

The reference case uses $T^{\text{DU}} = 0.5$ h for all products as a modeling assumption, not as a claim that every CAISO product has identical deployment and certification rules.

4.6. Retail costs

The TOU term intentionally retains the signed meter:

$$C_k^{\text{TOU}} = \sum_{t \in \mathcal{H}_k} C_t^{\text{TOU}} p_t^{\text{GI}} \Delta t. \quad (25)$$

Thus $p_t^{\text{GI}} < 0$ receives a retail-price export credit under the adopted tariff convention.

For demand category $q \in \{\text{PD}, \text{NCD}\}$, let $M_k^{q,\text{hist}}$ be the maximum eligible executed import before solve k . The avoidable increase is modeled with

$$z_k^q \geq 0, \quad z_k^q \geq p_t^{\text{GI}} - M_k^{q,\text{hist}}, \quad t \in \mathcal{H}_k \cap \mathcal{T}_q, \quad (26)$$

$$C_k^{q,\text{inc}} = C^q z_k^q. \quad (27)$$

After executing interval k ,

$$M_{k+1}^{q,\text{hist}} = \max\{M_k^{q,\text{hist}}, m_q(k) p_k^{\text{GI,executed}}\}. \quad (28)$$

The earlier expression $C^q \max\{M_k^{q,\text{hist}}, \max_t p_t^{\text{GI}}\}$ differs from (27) only by the constant $C^q M_k^{q,\text{hist}}$ within a solve:

$$C^q \max\{M, z\} = C^q M + C^q [z - M]_+. \quad (29)$$

Therefore both yield the same optimizer when they use the same meter variable and fixed historical threshold. Equation (27) is preferable for incremental accounting because it does not repeatedly charge sunk cost.

4.7. Wholesale revenue and objective

Wholesale energy and AS capacity revenue are

$$R_k^{\text{WM,energy}} = \sum_{t \in \mathcal{H}_k} \Delta t (\lambda_t^{\text{DA}} p_t^{\text{DA}} + \lambda_t^{\text{RT}} p_t^{\text{RT}}) + R_k^{\text{deploy}}, \quad (30)$$

$$R_k^{\text{WM,capacity}} = \sum_{t \in \mathcal{H}_k} \Delta t \sum_{x \in \mathcal{K}} \left(\rho_t^{x,\text{DA}} c_t^{x,\text{DA}} + \rho_t^{x,\text{RT}} c_t^{x,\text{RT}} \right), \quad (31)$$

$$R_k^{\text{EV}} = \sum_{t \in \mathcal{H}_k} C^{\text{EV}} p_t^{\text{EV}} \Delta t. \quad (32)$$

R_k^{deploy} prices the expected deployment energy in (19) at the RT energy price.

At MPC call k , the complete active-horizon problem is

$$\min_{\mathbf{x}_k \in \mathcal{X}(\mathcal{I}_k)} J_k = \sum_{t \in \mathcal{H}_k} C_t^{\text{TOU}} p_t^{\text{GI}} \Delta t + \sum_{q \in \{\text{PD}, \text{NCD}\}} C^q z_k^q - R_k^{\text{EV}} - R_k^{\text{WM,energy}} - R_k^{\text{WM,capacity}}, \quad (33)$$

subject to (1)–(24) and the corresponding binary linearizations. There is no DA-tracking penalty. The economic consequence of deviating from DA is already represented by the DA/RT two-settlement terms.

5. Information policies, horizons, and oracle logic

5.1. Perfect and Persistence EV information

Once a vehicle arrives, its realized session information is used. For sessions that have not arrived, Perfect supplies the realized future session set within the active horizon, while Persistence constructs the future set from recent charging behavior. Electricity prices are exogenous processed inputs in both cases; “Perfect” refers to EV information rather than to price forecasting.

5.2. Why 24-h Perfect need not be best

Let ω denote the complete realized billing-period data and $\mathcal{X}(\omega)$ the common full-period feasible set. A true oracle solves

$$J_{\text{oracle}}^*(\omega) = \min_{\mathbf{x} \in \mathcal{X}(\omega)} J_{\mathcal{T}}(\mathbf{x}; \omega). \quad (34)$$

For every feasible policy π ,

$$J_{\text{oracle}}^* \leq J^\pi \iff V_{\text{oracle}}^* \geq V^\pi, \quad V = -J. \quad (35)$$

The inequality direction differs for cost and revenue.

A 24-h MPC policy instead solves

$$\mathbf{x}_{k:k+H-1}^\pi \in \arg \min_{\mathbf{x}} \left[\sum_{\tau=k}^{k+H-1} \ell_\tau(\mathbf{x}_\tau; \mathcal{I}_k^\pi) + \widehat{V}_{k+H}(s_{k+H}) \right], \quad (36)$$

executes only $\mathbf{x}_k^\pi$, and repeats. The current controller does not know the exact continuation value V_{k+H}^* . Consequently, more accurate information inside the truncated horizon does not imply globally better closed-loop value [22].

The mechanism resembles a greedy algorithm. Each solve selects a locally attractive first action from incomplete future information. That action changes the next state and therefore the feasible set of every later solve. Four state channels matter here:

1. BESS SOC and daily throughput usage;
2. monthly PD and NCD thresholds;
3. EV energy remaining before departure; and
4. submitted DA energy and AS commitments that cannot be undone.

A Persistence forecast can accidentally preserve a more valuable out-of-horizon state than a Perfect 24-h solve. A 1% MIP gap can also select different near-optimal first actions and amplify their differences through the cascade, but this is a secondary numerical mechanism. The validation rule is therefore: the full-period oracle must dominate, while 24-h Perfect versus Persistence is an empirical policy comparison.

6. Case study and validation protocol

6.1. June 2025 configuration

The study covers 1–30 June 2025 at 15-min resolution. The BESS is rated at 250 kW and 332 kWh with SOC limits 0.05–0.95 and $\gamma = 0.9$. The aggregate EV charger limit is 6.6 kW per connected session. EV user revenue is \$0.40/kWh. The summer PD and NCD rates are \$3.05/kW and \$15.38/kW, respectively. Gurobi 13.0.1 is called through CVXPY 1.8.2 with a 1% relative MIP-gap target. The formal Rolling run uses no ordinary per-step time limit; abnormal-run protection is handled separately, and no TimeLimit event occurs.

All methods enter June from a common January–May baseline-state construction. The full-period oracle uses the same Perfect baseline used by Rolling Perfect, the same realized June input, the same physical constraints, and the same final SOC target. Shrinking results provide the complete five-method financial comparison, while detailed daily validation focuses on Rolling Persistence, Rolling Perfect, and the full-period oracle because these three cases isolate forecast and horizon effects under one controller structure.

6.2. Computational validation

The validation set contains 5,820 Rolling solves: two forecasts times 30 DA and 2,880 RT optimizations. Source identity, solver records, dispatch traces, daily audits, and vector figures are archived together to prevent numerical results from being separated from their implementation. All 19 acceptance checks pass. The accepted full-period oracle terminates normally at a reported 0.8571% gap. A separate 900-s attempt found a higher feasible incumbent but stopped at 1.2006%; it is retained as non-certified evidence and is not substituted into the consistently certified result table.

7. Results

7.1. Monthly five-method comparison

Table 4 and Fig. 3 report executed June results. Costs are stored as negative financial components, so

$$V = R^{\text{EV}} + R^{\text{WM}} + C^{\text{TOU}} + C^{\text{PD}} + C^{\text{NCD}}. \quad (37)$$

The full-period oracle exceeds the best 24-h policy, Shrinking Persistence, by \$1,579.73 or 14.38%, and exceeds Rolling Perfect by \$1,832.00. The benefit is not simply higher gross wholesale revenue. Relative to Rolling Perfect, the oracle gains \$217.66 in WM revenue and reduces TOU cost by \$1,895.85, while its NCD cost is \$445.91 larger. The net improvement is therefore a retail–wholesale scheduling effect across the complete billing period.

The 24-h ordering is not monotonic in forecast quality. Perfect exceeds Persistence by \$247.87 under Rolling MPC, whereas Persistence exceeds Perfect by \$432.75 under Shrinking MPC. Section 5 shows why this is permissible: forecast information, horizon, and path-dependent state jointly determine the executed policy. These values are empirical results, not a universal controller ranking.

7.2. Daily financial mechanism

Figure 4 separates daily attributed and cumulative value for the three detailed validation methods. Large negative daily values occur when a new monthly demand maximum is established; they should not be interpreted as an independent daily tariff. The cumulative plot is the appropriate billing-period comparison.

7.3. Resource and service-channel attribution

The net-value decomposition in Fig. 5 attributes each 24-h policy’s executed value to wholesale-market (WM) and non-wholesale (NWM) channels, EV and BESS resources, and energy and capacity services. This view reveals the mechanism hidden by a single wholesale-revenue total. WM energy and capacity contributions are positive for all four policies, while the NWM capacity attribution can be negative because market-oriented operation changes the retail-meter trajectory and monthly demand-charge exposure.

The daily decomposition is provided in Appendix Appendix B. It shows that most operating days make positive

Table 3: Formal validation hierarchy and acceptance criteria.

Layer	Criterion	Purpose
Solver completeness	30 DA + 2,880 RT solves per Rolling forecast; status optimal; gap $\leq 1\%$; no TimeLimit	Excludes silent truncation and missing intervals.
Physical identities	Meter, settlement, and actual up/down residuals $\leq 10^{-6}$ kW; actual AS $\geq -10^{-6}$ kW	Confirms the implemented dispatch matches the model.
EV service	Daily EV-energy spread across the three detailed scenarios $\leq 10^{-4}$ kWh	Prevents financial gains from unequal charging service.
Financial accounting	Daily component sum agrees with total within \$0.03	Allows independent cent rounding but no missing account.
Terminal state	Final billing-period SOC equals 0.5 within 10^{-6}	Prevents end-of-study energy borrowing.
Oracle consistency	identical Rolling-Perfect baseline and oracle value not below Rolling Perfect	Establishes a comparable full-period benchmark.
Graphical audit	PNG and vector PDF for every day	Links numerical checks to physical trajectories.

Table 4: Executed June 2025 value decomposition (US\$).

Method	Net value	WM revenue	TOU	PD	NCD	EV revenue
Full billing-period Perfect oracle	12,565.15	4,564.69	-10,404.95	-114.85	-4,873.66	23,393.92
Shrinking 24-h Persistence	10,985.42	4,424.62	-11,527.35	-527.59	-4,778.15	23,393.92
Rolling 24-h Perfect	10,733.15	4,347.03	-12,300.80	-279.25	-4,427.75	23,393.92
Shrinking 24-h Perfect	10,552.67	4,310.96	-12,413.73	-310.73	-4,427.75	23,393.92
Rolling 24-h Persistence	10,485.28	4,421.87	-11,092.27	-467.84	-5,770.41	23,393.92

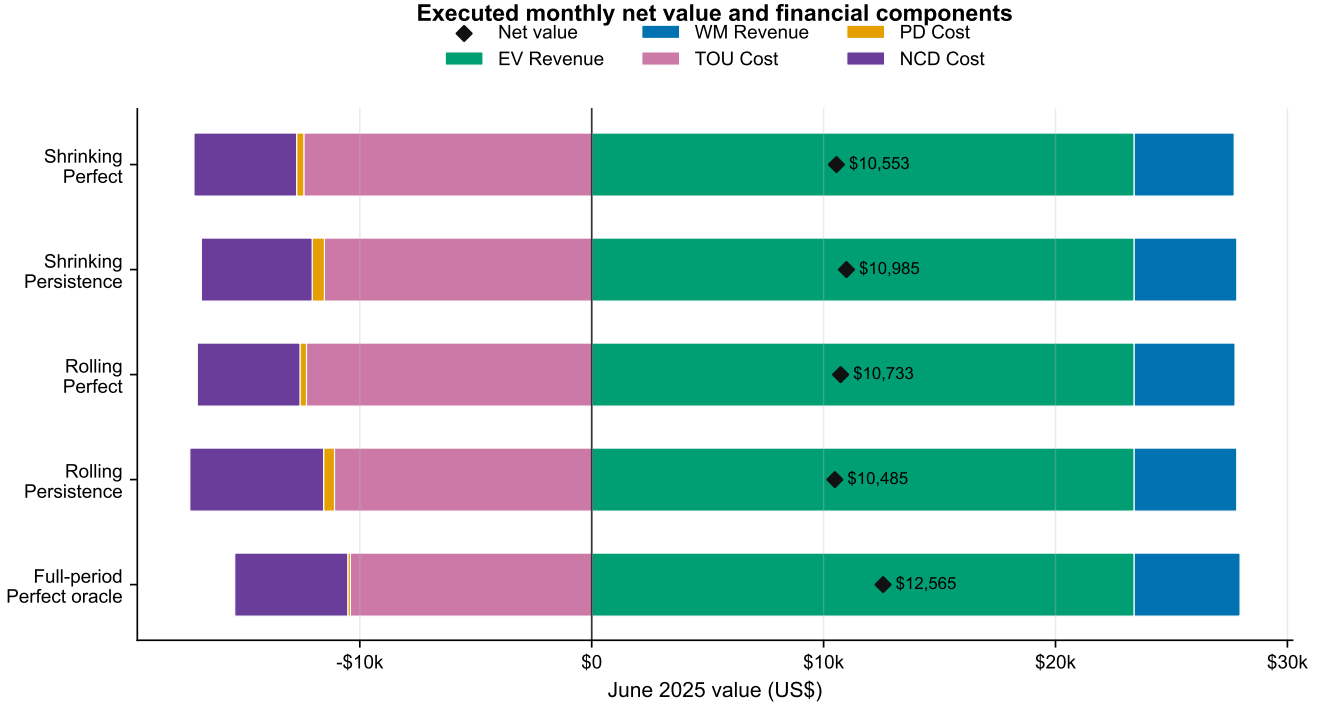


Figure 3: June 2025 executed value stack. The identical EV-revenue component confirms equal aggregate service; method differences arise from wholesale and retail interactions.

contributions, whereas a small number of days establish or increase the billing-period demand threshold and therefore carry large negative NWM capacity terms.

7.4. Numerical acceptance

The numerical audit is retained as reproducibility evidence rather than as a paper result figure. The largest formal meter residual is approximately 10^{-8} kW, the largest settlement-identity residual approximately 10^{-7} kW, and actual up/down-capability violations are of order 10^{-11} kW. All are below the 10^{-6} kW acceptance tolerance. Every Rolling solve terminates

with Gurobi status 2 and CVXPY status optimal; the maximum recorded relative gap is 0.999899%. Detailed daily residual, solver-gap, and stress-day plots are archived with the versioned validation outputs and are not presented as economic result figures.

8. Discussion

8.1. What is validated

The evidence supports correctness of the *reported implementation* within the stated tolerances: complete interval cov-

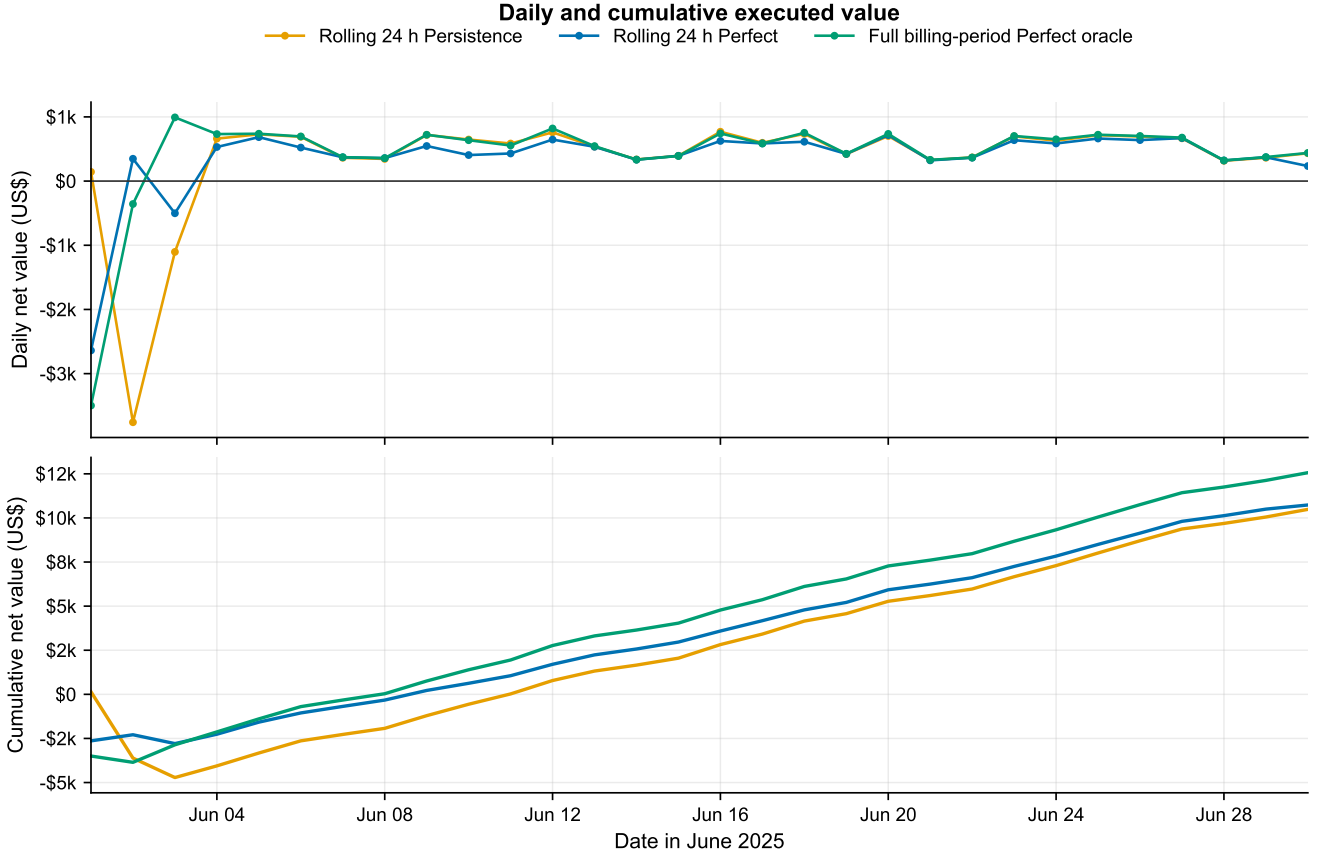


Figure 4: Daily and cumulative executed net value for Rolling Persistence, Rolling Perfect, and the full-period Perfect oracle.

erage; accepted solver status; meter, settlement, and capability identities; nonnegative actual AS; equal EV service; final SOC; and financial reconstruction. The full-period benchmark also confirms that the 24-h policies leave economically relevant value outside their look-ahead.

This does not prove that every modeling assumption is a market rule. Expected AS deployment factors, the common 0.5-h duration guard, the retail export credit, EV baseline construction, and known price trajectories remain explicit assumptions. The distinction is important for a rigorous journal claim: numerical implementation correctness and external market fidelity are related but separate questions.

8.2. Implications for forecast and sensitivity studies

The oracle gap identifies horizon value, not solely forecast value. Therefore, future forecast sensitivity should use matched initial SOC, monthly thresholds, and DA commitments when the purpose is to isolate RT forecast effects. For deployment studies, the more relevant metric is the closed-loop billing-period value of each policy, where state paths are allowed to diverge.

The next sensitivity analysis activates passive building load and PV through (12). Because these resources alter the signed meter but not the decision set, the base EV-BESS case remains intact and directly comparable. The main robustness questions are whether (i) oracle advantage, (ii) retail cost offsets to WM

revenue, and (iii) daily feasibility conclusions persist across PV/load scale and forecast cases.

8.3. Generalizable findings

Three findings define the paper’s contribution. First, the retail meter materially changes the value of a market-facing schedule, so gross WM revenue is not a reliable controller metric. Second, the full-period oracle improves value primarily by coordinating retail and wholesale states across the billing period, not by maximizing a single revenue account. Third, “Perfect forecast” is not synonymous with “upper bound” under truncated economic MPC; the upper bound requires the full feasible horizon and common boundary conditions.

9. Limitations and future work

The study uses one month of workplace EV data, and essentially no overnight EV carryover is represented. Prices are treated as known exogenous trajectories. EVs are aggregated rather than mapped to station-specific electrical limits. BESS degradation is represented only indirectly through efficiency and throughput constraints. The wholesale model uses expected AS deployment and does not reproduce regulation mileage, performance scores, no-pay charges, telemetry, minimum bid sizes, scheduling-coordinator requirements, or

Executed profit attribution by resource and service channel

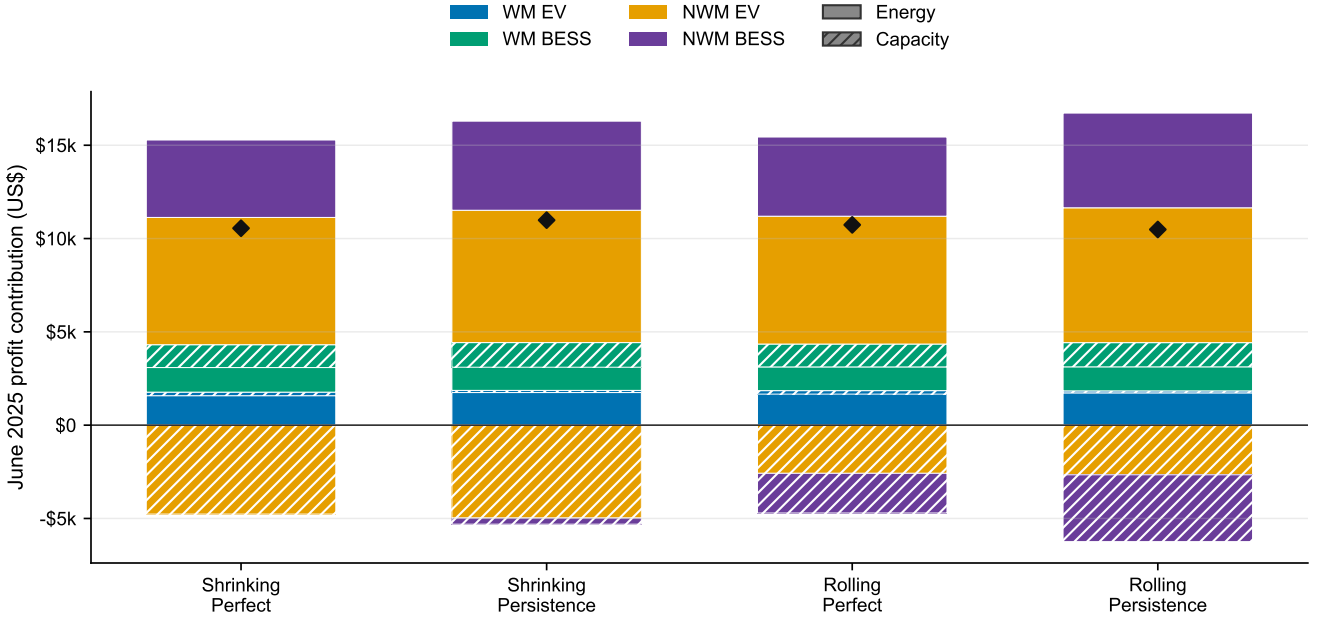


Figure 5: June 2025 executed-value attribution for the four 24-h policies. Color distinguishes WM/NWM and EV/BESS channels; hatching distinguishes energy and capacity components. Black diamonds show total net value.

every CAISO settlement rule. The full-period benchmark is solved to a 1% MIP gap, not proven to exact global optimality.

Future work should: (i) repeat the validated base case across additional seasons; (ii) add the passive PV/building sensitivity without replacing the base case; (iii) introduce product-specific AS duration and activation models; (iv) examine alternative EV baselines; (v) test forecast models beyond Persistence using matched-state experiments and closed-loop billing-period evaluation; and (vi) model station/network topology when site-level data become available.

10. Conclusion

An EV-BESS portfolio behind a campus meter must be evaluated as one coupled physical and financial system. The proposed framework preserves mandatory EV service, BESS feasibility, signed retail-meter accounting, fixed DA commitments, signed RT adjustments, actual nonnegative AS, and executed first-step revenue reconstruction. In the validated June 2025 case, the full-period Perfect benchmark produces \$12,565.15, 14.38% more than the best 24-h policy. All formal solver, physical, financial, and graphical checks pass. The result establishes a defensible benchmark and a reproducible validation standard for the next PV/building and forecast-robustness analyses.

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Appendix A. Complete financial objective and accounting identity

For a full evaluation set $\mathcal{T}$, the oracle objective corresponding to (33) is

$$\begin{aligned}
 \min_{\mathbf{x} \in \mathcal{X}(\omega)} J_{\mathcal{T}} = & \sum_{t \in \mathcal{T}} C_t^{\text{TOU}} p_t^{\text{GI}} \Delta t + C^{\text{PD}} z^{\text{PD}} + C^{\text{NCD}} z^{\text{NCD}} \\
 & - \sum_{t \in \mathcal{T}} C^{\text{EV}} p_t^{\text{EV}} \Delta t \\
 & - \sum_{t \in \mathcal{T}} \Delta t (\lambda_t^{\text{DA}} p_t^{\text{DA}} + \lambda_t^{\text{RT}} p_t^{\text{RT}}) \\
 & - \sum_{t \in \mathcal{T}} \sum_{x \in \mathcal{K}} \Delta t \rho_t^{x, \text{DA}} c_t^{x, \text{DA}} \\
 & - \sum_{t \in \mathcal{T}} \sum_{x \in \mathcal{K}} \Delta t \rho_t^{x, \text{RT}} c_t^{x, \text{RT}} - R^{\text{deploy}}. \quad (\text{A.1})
 \end{aligned}$$

The financial post-processor stores costs with negative signs. After independent cent rounding by day, the validation accepts

$$|V - (R^{\text{EV}} + R^{\text{WM}} + C^{\text{TOU}} + C^{\text{PD}} + C^{\text{NCD}})| \leq \$0.03. \quad (\text{A.2})$$

Appendix B. Daily resource and service-channel attribution

Daily executed profit attribution

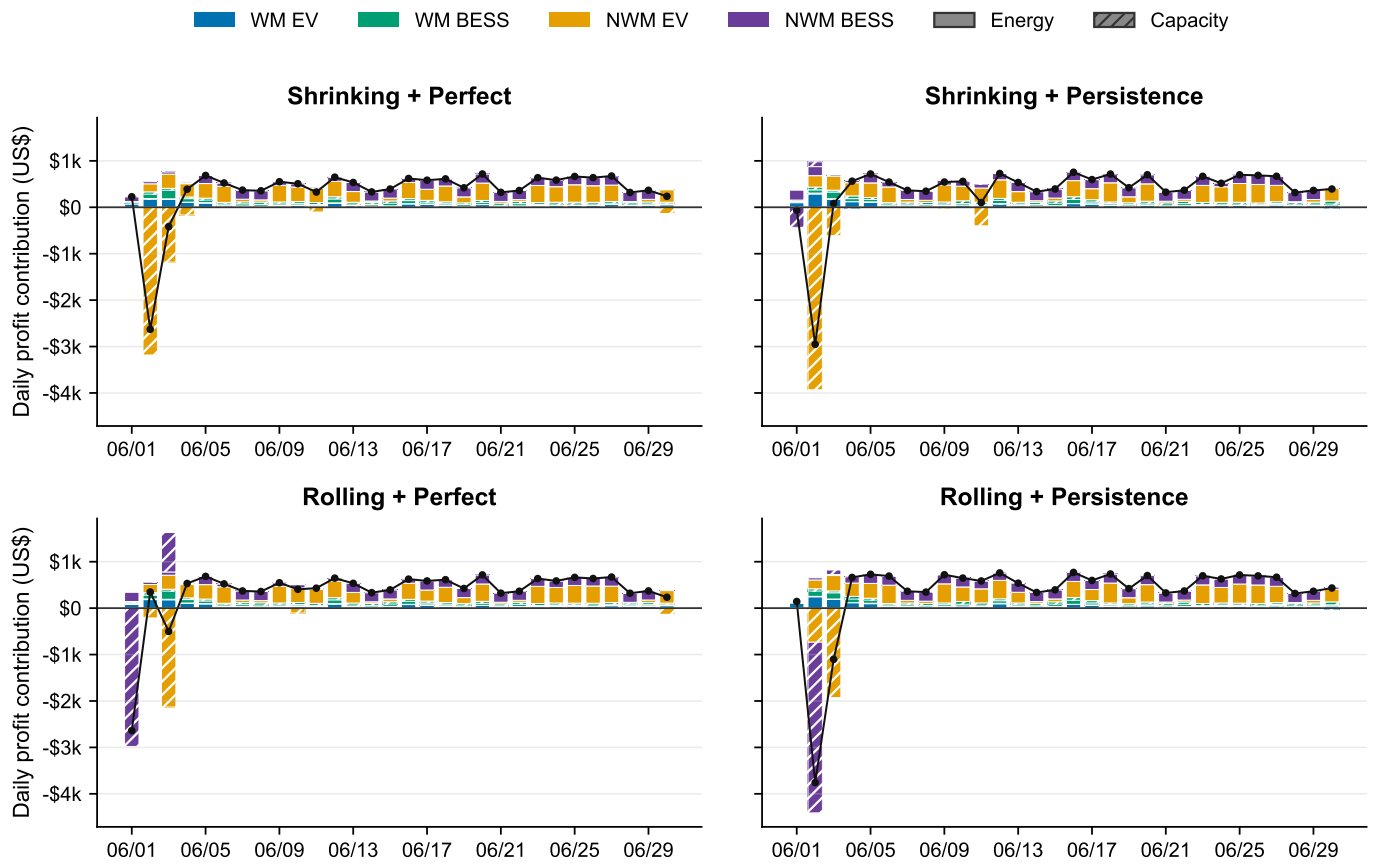


Figure B.6: Daily executed-value attribution for the four 24-h policies. The black trace is daily net value; stacked bars separate WM/NWM, EV/BESS, and energy/capacity contributions.

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